

Question	Answer	Marks
8(a)	(minimum) energy required to separate the nucleons (of a nucleus)	M1
	to infinity	A1
8(b)(i)	4	A1

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8(b)(ii)	energy = $(142 \times 8.37) + (90 \times 8.72) - (235 \times 7.59)$	C1
	= 190 MeV	A1
8(b)(iii)	either it has too many neutrons (for the number of protons) or its neutron to proton ratio is too high	B1
8(b)(iv)	(when $t = 6.0$ s), $N / N_0 = 1 / 32$	C1
	either $(1 / 32) = \exp(-\ln 2 \times 6.0 / t_{1/2})$	C1
	$t_{1/2} = 1.2$ s	A1
	or $32 / 2^n = 1$ so $n = 5$ (half-lives)	(C1)
	$t_{1/2} = 6.0 / 5$ = 1.2 s	(A1)

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